# Abhishek Kumar Singh

*SDE 2 – Systems, Performance & Backend*

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## Summary

Software Engineer with 3+ years building low-latency, high-throughput systems in **C++** and **Python** for safety-critical, real-time environments. Improved production throughput by **30%** and cut system failures by **90%** through cache-aware multithreading and hard concurrency debugging on live streaming pipelines. Comfortable across the stack: RTOS-level systems programming, distributed/lock-free architectures processing **1M+ events/sec**, and REST-integrated, database-backed services. Ranked in the top 0.5% globally in Google Code Jam and top 1% in Meta Hacker Cup.

## Experience

**Boeing** Bangalore, India

*Software Development Engineer II (SDE 2)* *Sep 2023 – Present*

- Designed and built low-latency, multithreaded services in **C++17** on **RTOS** processing high-volume real-time data streams for Flight Deck Video Display systems, powering live cockpit video pipelines.
- Improved system throughput by **30%** by redesigning core data pipelines with cache-aware multithreading and memory-access optimizations.
- Root-caused and resolved critical concurrency defects (race conditions, deadlocks) causing production freezes, reducing system failures by **90%** and eliminating recurring field escapes.
- Built test and validation frameworks for data-integrity checks and stale-data detection, hardening pipeline reliability under DO-178-aligned certification standards; partnered with QA and systems integration teams to ship certifiable releases.

**Goldman Sachs** Remote

*Cybersecurity Virtual Intern* *May 2023 – Jun 2023*

- Configured enterprise security tooling (firewalls, IDS/IPS, VPNs) and evaluated cryptographic protocols used in financial infrastructure as part of a broader security risk analysis.

**JP Morgan Chase & Co.** Remote

*Quantitative Developer Virtual Intern* *Sep 2022 – Nov 2022*

- Built the backend for a live financial data dashboard in **Python** and **JavaScript**, integrating real-time market data feeds and serving them to analyst-facing views.
- Automated anomaly-detection data pipelines to flag irregularities across large-scale financial datasets, reducing manual review effort.

## Projects

**Trading Simulator** | *C++, Lock-Free Programming, Linux, WebSockets* GitHub | Live Demo

- Built a multithreaded, sharded order-matching engine in C++ with a custom lock-free MPMC queue, sustaining **1.6–2.4M orders/sec** on a real multi-core Linux benchmark, with O(1) order cancellation via an iterator-stable index (`std::list` + `unordered_map`).
- Diagnosed a bufferbloat-style latency/throughput tradeoff via hypothesis-driven benchmarking, then tuned queue depth to cut p50 latency **~210x (177ms → 0.84ms)** with no measurable throughput cost.
- Shipped three interfaces on the same core engine: a benchmark CLI with an auto-generated results dashboard, a Qt desktop GUI, and a hand-rolled WebSocket server (RFC 6455 from scratch) powering a live-hosted web demo.

**Real-Time Video Fault Detection Pipeline** | *C++, Linux, Multithreading*

- Designed a checksum- and timestamp-based stale/frozen-frame detection mechanism achieving deterministic sub-100ms fault detection latency in live video streams, with diagnostic logging to isolate root causes.
- Built a custom regression testbench and validation framework for data-integrity and stale-data checks, hardening pipeline reliability under DO-178-aligned certification standards.

**CodeBuddy** | *Node.js, Express, SQLite, REST APIs* GitHub | Live Demo

- Built a full-stack, multi-language (12 languages) online code runner: a vanilla JS/CodeMirror frontend and a Node.js/Express REST API backend proxying the Judge0 execution engine, deployed live on Render.
- Designed a SQLite-backed persistence layer with indexed queries tracking every run (language, status, execution time), powering a live run-history panel and per-language success-rate stats.
- Implemented API-layer rate limiting and input validation to protect the shared execution backend; load-tested to confirm exactly 20 requests/min allowed before **429** responses.

## Technical Skills

**Languages**: C++, Python, JavaScript, Go, SQL
**Systems & Performance**: Multithreading, Concurrency, Lock-Free Programming, Memory Optimization, RTOS, High-Throughput/Low-Latency Pipelines
**Backend & Data**: REST APIs, Distributed Systems, PostgreSQL, SQL, Real-Time Data Pipelines
**Infrastructure**: Linux, Docker, Cloud Computing, Storage Architecture
**Practices & Tools**: Git, Software Test Engineering, Performance Profiling & Analysis, Debugging

## Education

**Kamla Nehru Institute of Technology**
*Bachelor of Technology, Computer Science – GPA: 8.2/10* *Jul 2019 – May 2023*

## Achievements

- **Google Code Jam** – Ranked in the top 0.5% globally out of 400,000+ participants.
- **Meta Hacker Cup** – Ranked in the top 1% worldwide.
- **Google Hash Code** – Achieved a global rank of 2,508.
- **CodeChef** – 4-star rated competitive programmer (max rating 1908).